Problem statement – Underlying reason for development of this repository

To evaluate thermal comfort in built environments PMV of ASHRAE 55 is widely used. The comfort assessment is made using the power of computational fluid dynamics. The most common way is to run a CFD simulation case with the following models:

i. Turbulence, k-omega SST
ii. Energy
iii. Species
iv. Radiation – DO (discrete ordinates)

The converged result which we get from running the above case gives us the values of dry bulb temperature, velocity, RH and MRT. These values along with clothing and metabolic factors are then used to calculate PMV. This evaluation tool (PMV) works perfectly well for indoor steady state environments. The CFD engineer can also employ a UDF in Fluent Solver for getting PMV/PPD contours which are based on the six factors (DBT, V, RH, MRT, CLO, M). For PMV/PPD, the method is straightforward, simple and accurate.

Whereas, for outdoor unsteady and abrupt environment, PMV cannot be used as an evaluating tool [1] [2] [3] [4] [5] [6]. For outdoor comfort the best and validated method is to use the Standard Effective Temperature (SET) model which is also known as Gagge 2-node model of thermal comfort. This model considers human thermoregulation by splitting the human body into two nodes, i.e., skin node and core node. The core node produces metabolic heat and maintains temperature of 37 degrees Celsius. The skin node exchanges heat with the environment by evaporation, convection etc. Now, to model this in CFD in the conventional way using UDF, there are many issues related to its stability, validation and accuracy. The main issue is numerical stability. The following are the issues faced:

1. The user defined forces UDFs make the solution diverge.
2. Since SET is calculated through iterative process the iterative process also slows down the solver speed if the solver is stable.
3. UDF for SET is more prone to error which will give misleading results.
4. The UDF code should be validated, which is very difficult.
5. The bisection solver causes convergence failures under outdoor conditions. Without throwing errors it will give wrong results.
6. There is floating point overflow in sweating equation of SET. This requires adding limits to the c code.
7. The SET from UDF never matches the SET from CBE tool [7].
8. I, personally, also felt that there was no proper heat balance between 2-node of human model and the outdoor environment in the whole CFD domain. It was very difficult to create a qualifying mesh.

9. The only way to get acceptable SET value through UDF was by making approximations in the UDF code. These approximations break the Gagge's 2-node theory.

Solution

The solution to the above problem is the following:

1. Use the tools which are best in their own domain. For example, use Ansys Fluent (or OpenFoam) for converging the solution with accuracy. This will provide the final values of DBT, Velocity, RH and MRT.
2. Use pythermalcomfort library to evaluate thermal comfort. Use the above four values with clothing factor and metabolic activity to calculate SET using industrial standard and validated tool – CBE thermal comfort tool [7] ***(for automation we will use the power of industrial standard pythermalcomfort library [8]. See the paragraphs below)***

When the CFD model or geometry is small and where the regions of interest are less the DBT, V, MRT, RH can be extracted easily and then fed to CBE tool for calculating SET. This is easy, quick and simple when the CFD domain / geometry is small.

Another problem arises when the geometry is very big and there are 100s of regions of interest where we need the final value of SET. Extracting the values manually and calculating SET will take very long time and using this simple procedure we will only get the value of SET in number with no contours of SET to plot in our CFD domain. To view the contours of SET and to automate everything, for very large scale mega CFD projects, the procedure below is developed as follows (the steps below are for advanced Fluent 2026R1 users, for detailed steps see the readme file in repository):

1. Create a CFD model with wind tunnel. Generate good quality mesh. Setup Fluent solver with all the required equations i.e., Turbulence (kw-SST), Energy, Species (for RH) and Discrete Ordinates (DO) for MRT.
2. Run the simulation. Monitor convergence plots. Stop when deemed converged.
3. Start the *grpc-server* in Fluent console to communicate with *PyFluent* library. Connect the Fluent session with PyFluent.
4. You should have all the python dependencies installed including the precompiled Google *protobuf* library.
5. Extract the CFD domain data with DBT, RH, V, MRT. All with the coordinates of cells and cell ids.
6. Calculate the SET using the four extracted values from CFD + Clo and M.
7. Match the SET and cell id with cell coordinates.
8. Create UDM 0 in Fluent.
9. Rename UDM 0 to set-comfort by UDF.
10. Create a UDF to inject SET values in cell coordinates.

11. View the contours of SET in your full CFD domain in the Fluent Solver which is also exportable to CFD-Post for more post processing.
12. **For detailed steps from scratch refer to the readme file in repository.**

Important notes:

1. *The steps from 5 to 11 are automated by using the scripts in my repository. The scripts are light weight such that they are created considering 100+ million mesh cells in mind. The scripts process data in chunks and inject data in chunks so that the RAM consumption does not hit 100% of installed RAM while the Fluent solver is open. For calculating SET the script does not store the data of 50 or 100 million cells in RAM rather it writes a new temporary file in the directory, processes the data, then deletes it. This was done to prevent huge RAM consumption.*

2. *The scripts are tested on 422,000 mesh cells with 32 core parallel processing. These scripts inject the new contour parallelly (on 32 nodes or any number of cores you have) without any MPI errors. Extraction/injection in serial or single code is easy. The challenges lie when extraction/injection are done in parallel mode. All these challenges were resolved for use in parallel mode.*

3. *The injection of SET values in cells is carried out based on the coordinates of cells. The extraction script extracts the data with local cell IDs. In parallel mode each partition/core has its own local cell ID which depends on the number of cores the user is using. This will create conflict during injecting as the cell ids will be repeated for each partition. Also, IDs get changed whenever a new Fluent session is opened with same mesh which will cause wrong data injection in wrong cells. To overcome this, in the script, the injection is done based on the coordinates of each cell in the whole domain. The coordinates remain the same for the same mesh irrespective of the number of times a case/mesh is opened in Fluent solver. Fluent solver repartitions the grid with new cell ids, but the coordinates of each cell remain the same unless a new mesh is generated. Therefore, injecting SET data based on the location of cell in X Y Z coordinates is the correct and proper way.*

4. *These scripts were generated with the help of Gemini AI by constant and repeated prompts including all the troubleshooting. Hence, now, after 15th May 2026, Gemini AI is trained to answer and generate the code for PyFluent with Fluent 2026 R1 solver.*

5. *The main aim of this project was to reduce the time to CFD results for very large scale outdoor thermal comfort projects.*

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